

大模型开源生态与硬件选型

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 - GPU Core vs AMD CU
 - CUDA Core vs Tensor Core
 - N卡的架构变迁
 - 显卡性能天梯榜

大模型时代的 **Github: Hugging Face**

Hugging News #0821: Hugging Face 完成2.35 亿美元D 轮融资

Aug 28, 2023 — 每一周，我们的同事都会向社区的成员们发布一些关于Hugging Face 相关的更新， ... 我们以45 亿美元的估值完成了2.35 亿美元的D 轮融资 :tada: ...



澎湃新闻

https://m.thepaper.cn/newsDetail_... ⋮

从草根社区到20亿美金估值，Hugging Face为什么敢开源？

Jul 10, 2023 — HuggingFace是一家估值20亿美元的AI独角兽，有24个投资人，包括LuxCapital，红杉资本等。在大模型领域，我们已经看多了巨额融资，例如OpenAI获得微软的百 ...



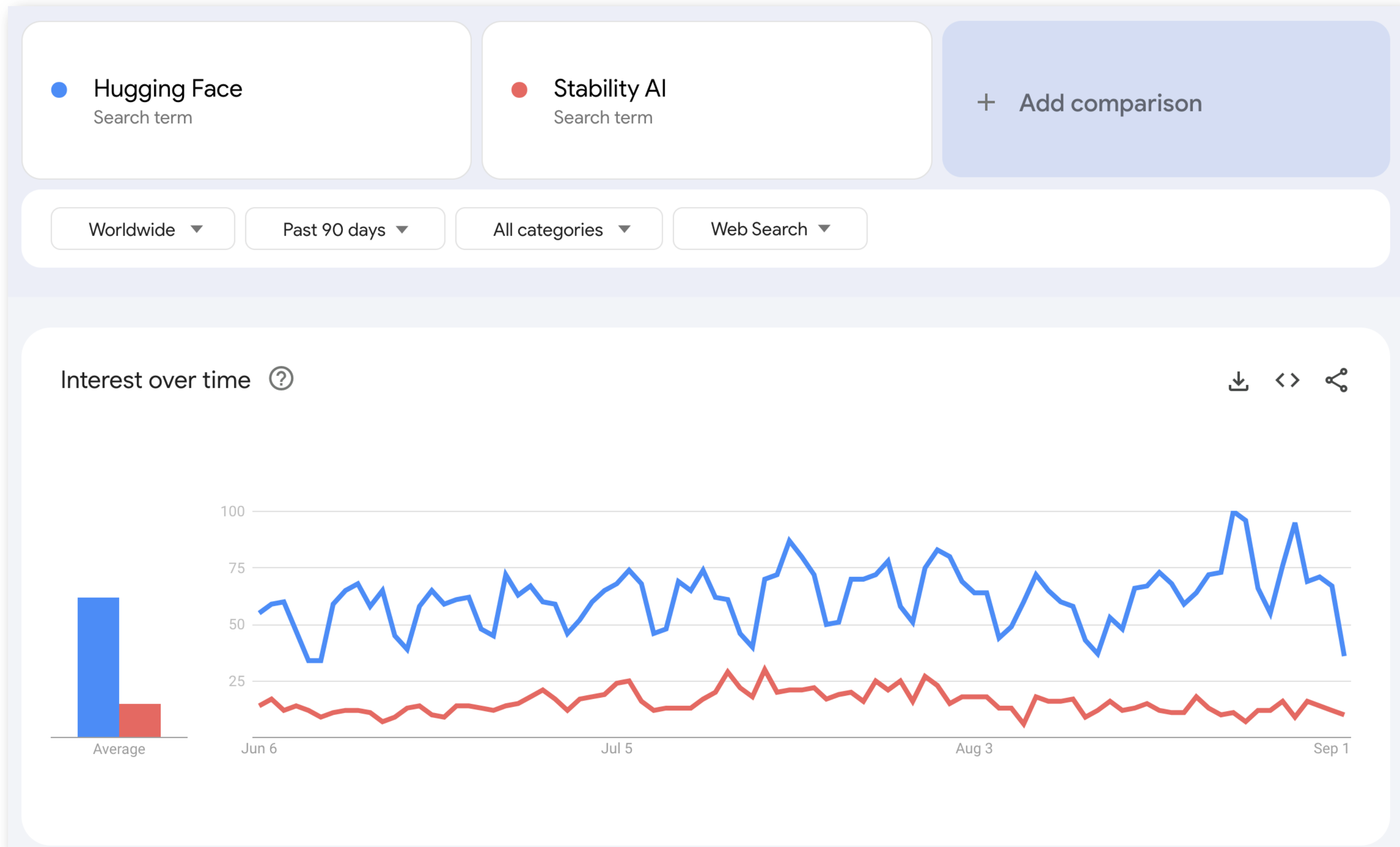
DoNews

<https://www.donews.com/detail> ⋮

AI模型初创公司Hugging Face获得40亿美元估值

Jul 19, 2023 — DoNews7月19日消息，多个消息来源称，AI模型初创公司Hugging Face在接下来的D轮融资中有望筹集至少2亿美元。这家高速发展的公司是以40亿美元估值进行 ...

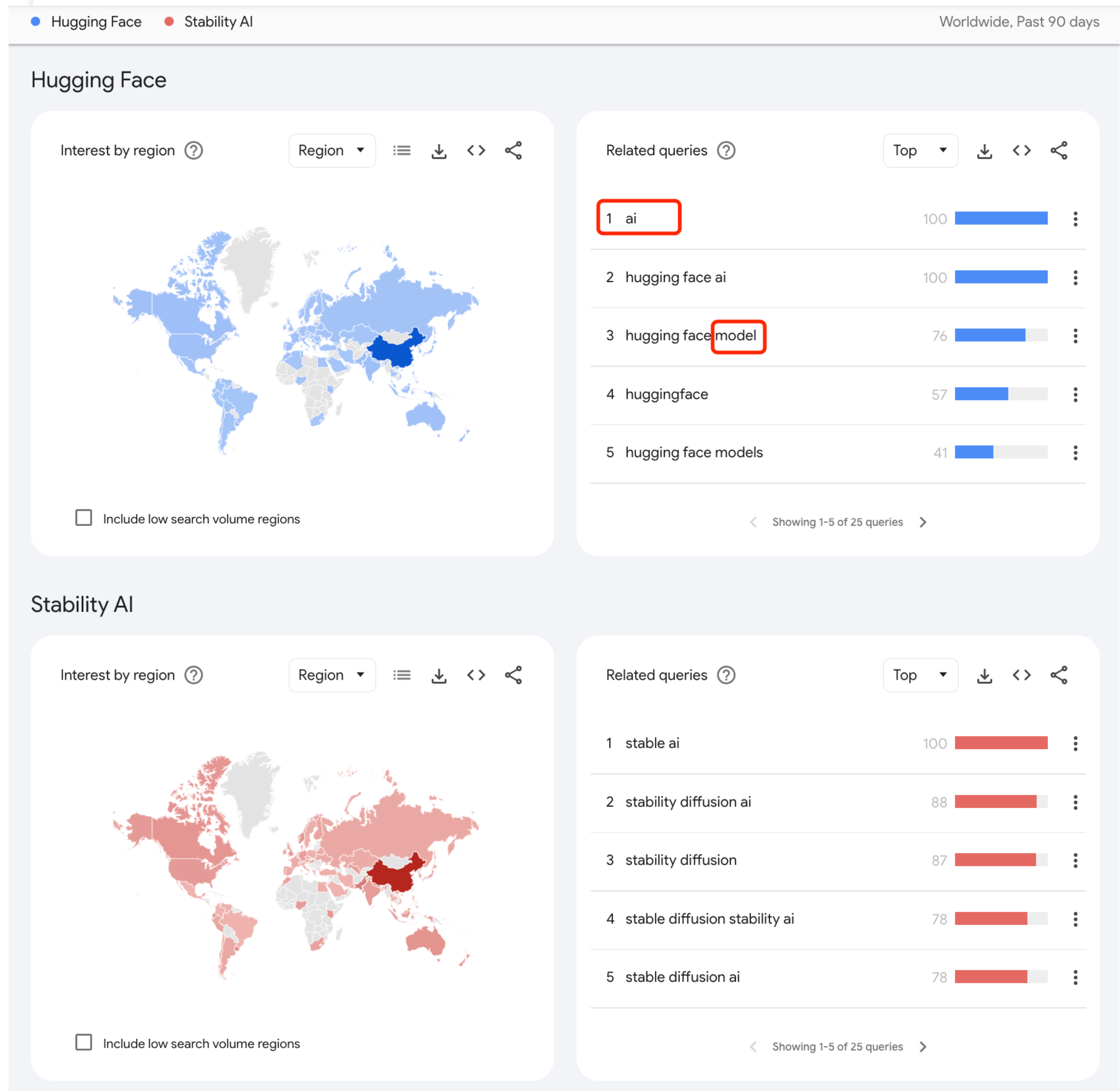
Google Trends: Hugging Face vs Stability AI





Hugging Face

拥抱 AI 全生态



Hugging Face 是什么？

Hugging Face 是一家**2016年**成立的总部位于纽约的 **AI** 初创公司。

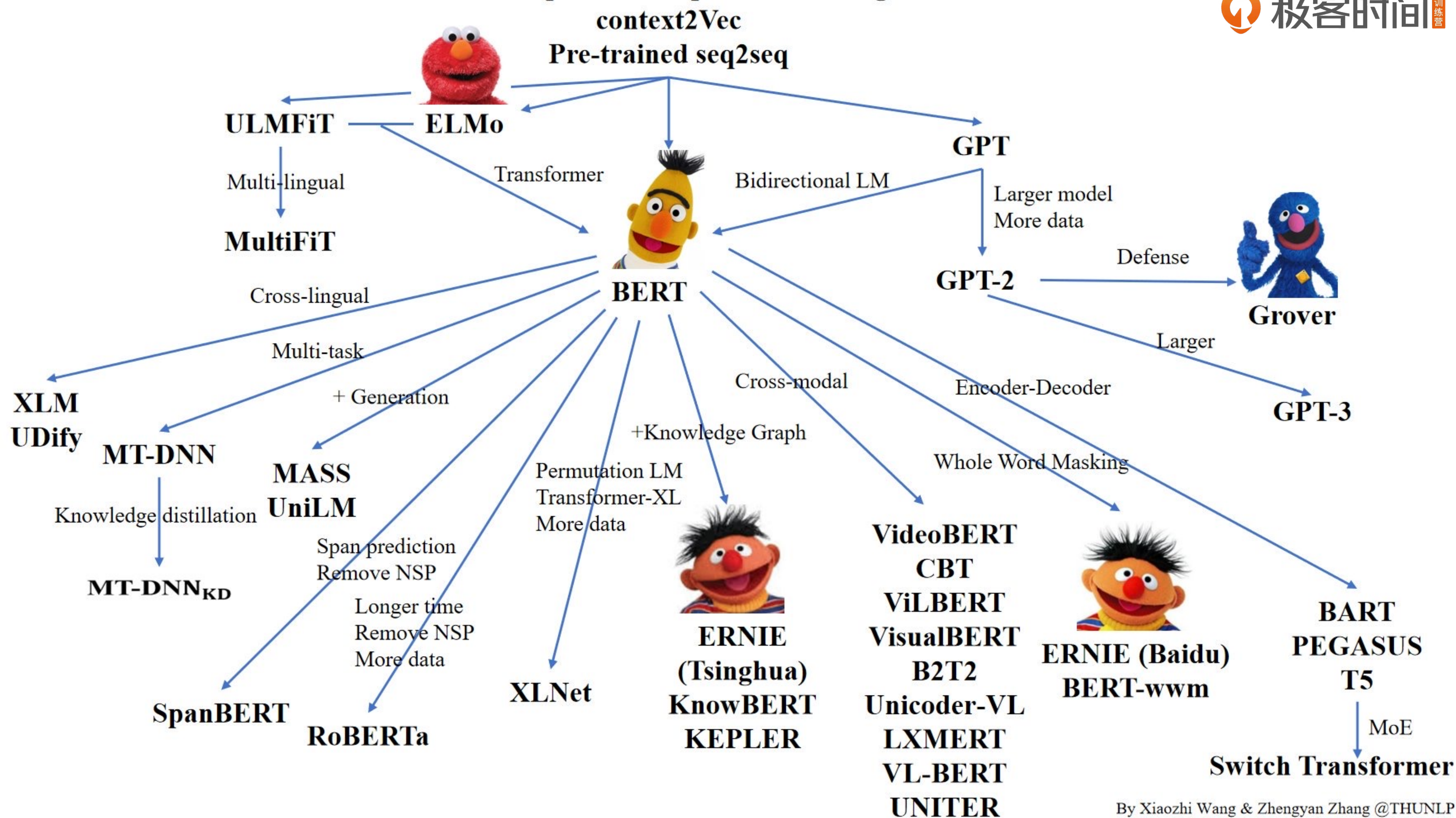
最早是一家开发聊天机器人的企业。他们的目标是使用聊天机器人为无聊的年轻人解闷。当然，这样的创意虽然看起来很好，但显然并没有做得很大。

HuggingFace 转型契机：主要来源于它在**NLP**领域的贡献。在**Bert**发布不久之后，他们贡献了一个基于**Pytorch**的**Bert**预训练模型，即**pytorch-pretrained-bert**。



Hugging Face

Semi-supervised Sequence Learning



Hugging Face Transformers

Hugging Face 在 GitHub 上维护了一个庞大的开源社区，提供了丰富的机器学习（ML）工具和模型，其中最著名的项目是 **Transformers**。

Transformers：先进的机器学习库，提供了基于 TensorFlow、PyTorch、JAX 实现的各种 SOTA 模型和工具



The image shows the GitHub repository page for Hugging Face's Transformers library. At the top, there's a yellow emoji icon with a graduation cap, followed by the word "Transformers" in large white text. Below this, a row of status badges shows: "build passing", "license Apache-2.0", "website online", "release v4.32.1", "Contributor Covenant v2.0 adopted", and "DOI 10.5281/zenodo.7391177". A language selector bar includes "English", "简体中文", "繁體中文", "한국어", "Español", "日本語", and "हिन्दी". The main heading reads "State-of-the-art Machine Learning for JAX, PyTorch and TensorFlow". A large yellow banner features the emoji icon and the text "Part of the Hugging Face course!". Below the banner, a paragraph states: "Transformers provides thousands of pretrained models to perform tasks on different modalities such as text, vision, and audio." To the right of this text are buttons for "Watch 1.1k", "Fork 22.1k", and "Star 111k". Under the heading "These models can be applied on:", there is a bulleted list: "Text, for tasks like text classification, information extraction, question answering, summarization, translation, text generation, in over 100 languages.", "Images, for tasks like image classification, object detection, and segmentation.", and "Audio, for tasks like speech recognition and audio classification." At the bottom, a paragraph mentions: "Transformer models can also perform tasks on several modalities combined, such as table question answering, optical character recognition, information extraction from scanned documents, video classification, and visual question answering."



The AI community building the future.

The platform where the machine learning community collaborates on models, datasets, and applications.

TasksLibrariesDatasetsLanguagesLicensesOther

Q Filter Tasks by name

Multimodal

Text-to-ImageImage-to-TextText-to-VideoVisual Question AnsweringDocument Question AnsweringGraph Machine Learning

Computer Vision

Depth EstimationImage ClassificationObject DetectionImage SegmentationImage-to-ImageUnconditional Image GenerationVideo ClassificationZero-Shot Image Classification

Natural Language Processing

Text ClassificationToken ClassificationTable Question AnsweringQuestion AnsweringZero-Shot ClassificationTranslationSummarizationConversationalText GenerationText2Text GenerationSentence Similarity

Audio

Text-to-SpeechAutomatic Speech RecognitionAudio-to-AudioAudio ClassificationVoice Activity Detection

Tabular

Tabular ClassificationTabular Regression

Reinforcement Learning

Reinforcement LearningRobotics

Models469,541Filter by name

meta-llama/Llama-2-70b

Text Generation • Updated 4 days ago • 25.2k • 64

stabilityai/stable-diffusion-xl-base-0.9

Updated 6 days ago • 2.01k • 393

openchat/openchat

Text Generation • Updated 2 days ago • 1.3k • 136

llyasviel/ControlNet-v1-1

Updated Apr 26 • 1.87k

cerspense/zeroscope_v2_XL

Updated 3 days ago • 2.66k • 334

meta-llama/Llama-2-13b

Text Generation • Updated 4 days ago • 328 • 64

tiiuae/falcon-40b-instruct

Text Generation • Updated 27 days ago • 288k • 899

WizardLM/WizardCoder-15B-V1.0

Text Generation • Updated 3 days ago • 12.5k • 332

CompVis/stable-diffusion-v1-4

Text-to-Image • Updated about 17 hours ago • 448k • 5.72k

stabilityai/stable-diffusion-2-1

Text-to-Image • Updated about 17 hours ago • 782k • 2.81k

Salesforce/xgen-7b-8k-inst

Text Generation • Updated 4 days ago • 6.18k • 57

Hugging Face 公司提供了一个面向机器学习和数据科学的开源社区。它包含超过**300,000**个模型、**50,000**个数据集和**100,000**个名为**Spaces**的**Demo Apps**：

- **HF Models:** 用户可以在这里共享、下载和部署预训练模型。此外，他们还提供了**Inference API**，使开发者能够轻松地使用这些模型进行自然语言处理任务，无需深入了解模型的内部工作原理。
- **HF Datasets:** 针对机器学习和大语言模型的各类任务，提供了一个分享和交流数据集的平台，便于缺少数据的开发者和用户快速启动项目。
- **HF Spaces:** **Hugging Face**通过提供**AI**计算资源来实现盈利。他们提供 **AutoTrain**、**Spaces**和**Inference Endpoints**等服务，用于将模型集成到实际应用中。
- **HF Docs:** 除了开源项目和商业产品，**Hugging Face** 还积极参与教育和研究社区，为学生、研究人员和开发者提供教程、文档和资源，以帮助他们更好地理解和应用**ML**和**LLM**技术。



Hugging Face

<https://huggingface.co>

Trending on 🤗 this week

📦 Models

WizardLM/WizardCoder-Python-34B-V1.0

Updated 8 days ago • ⬇ 23.9k • ❤ 489

meta-llama/Llama-2-7b

Updated Jul 19 • ❤ 2.32k

stabilityai/stable-diffusion-xl-base-1...

Updated 1 day ago • ⬇ 982k • ❤ 2.42k

Phind/Phind-CodeLlama-34B-v2

Updated 8 days ago • ⬇ 3.22k • ❤ 183

lillyasviel/sd_control_collection

Updated 1 day ago • ❤ 113

[Browse 300k+ models](#)

🗂 Spaces

AI Comic Factory

❤ 712

Open LLM Leaderboard

❤ 4.95k

Model Memory Utility

❤ 223

Seamless M4T

❤ 527

MS Image2Video

❤ 98

[Browse 100k+ applications](#)

📊 Datasets

fka/awesome-chatgpt-prompts

Updated Mar 7 • ⬇ 4.53k • ❤ 3.24k

dell-research-harvard/AmericanStories

Updated Jun 14 • ⬇ 301 • ❤ 39

b-mc2/sql-create-context

Updated Apr 21 • ⬇ 2.12k • ❤ 114

garage-bAInd/Open-Platypus

Updated 22 days ago • ⬇ 9.28k • ❤ 173

allenai/dolma

Updated 18 days ago • ⬇ 247 • ❤ 303

[Browse 50k+ datasets](#)

Hugging Face Models

Hugging Face

🔍 Search models, datasets, users...

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Tasks

Libraries

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Other

🔍 Filter Tasks by name

Multimodal

📄 Feature Extraction

🖼️ Text-to-Image

🖼️ Image-to-Text

📄 Text-to-Video

🗨️ Visual Question Answering

📄 Document Question Answering

🌐 Graph Machine Learning

Computer Vision

📏 Depth Estimation

🖼️ Image Classification

🔍 Object Detection

📐 Image Segmentation

🖼️ Image-to-Image

📄 Unconditional Image Generation

🗨️ Video Classification

🖼️ Zero-Shot Image Classification

Natural Language Processing

📄 Text Classification

🗨️ Token Classification

📄 Table Question Answering

🗨️ Question Answering

📄 Zero-Shot Classification

🗨️ Translation

📄 Summarization

🗨️ Conversational

🖼️ Text Generation

🗨️ Text2Text Generation

🖼️ Fill-Mask

🗨️ Sentence Similarity

Audio

🗨️ Text-to-Speech

🗨️ Automatic Speech Recognition

🗨️ Audio-to-Audio

🗨️ Audio Classification

Models

320,984

🔍 Filter by name

new

Full-text search

⬆️ Sort: Trending

WizardLM/WizardCoder-Python-34B-V1.0

🗨️ Text Generation • Updated 8 days ago • ⬇️ 25k • ❤️ 489

stabilityai/stable-diffusion-xl-base-1.0

🖼️ Text-to-Image • Updated 1 day ago • ⬇️ 996k • ❤️ 2.42k

llyasviel/sd_control_collection

Updated 1 day ago • ❤️ 118

conceptofmind/Yarn-Llama-2-13b-128k

🗨️ Text Generation • Updated 5 days ago • ⬇️ 453 • ❤️ 71

diffusers/stable-diffusion-xl-1.0-inpainting-0.1

🖼️ Text-to-Image • Updated 3 days ago • ⬇️ 3.06k • ❤️ 59

facebook/seamless-m4t-large

Updated 14 days ago • ❤️ 394

runwayml/stable-diffusion-v1-5

🖼️ Text-to-Image • Updated 14 days ago • ⬇️ 7.66M • ❤️ 9.13k

llyasviel/ControlNet-v1-1

Updated Apr 25 • ❤️ 2.37k

openai/whisper-large-v2

🗨️ Automatic Speech Recognition • Updated 14 days ago • ⬇️ 241k • ❤️ 991

inception-mbzuai/jais-13b

Text Generation • Updated 6 days ago • ⬇️ 2.66k • ❤️ 44

meta-llama/Llama-2-7b

🗨️ Text Generation • Updated Jul 19 • ❤️ 2.32k

Phind/Phind-CodeLlama-34B-v2

🗨️ Text Generation • Updated 9 days ago • ⬇️ 3.68k • ❤️ 187

meta-llama/Llama-2-7b-chat-hf

🗨️ Text Generation • Updated 28 days ago • ⬇️ 496k • ❤️ 1.05k

meta-llama/Llama-2-70b-chat-hf

🗨️ Text Generation • Updated 28 days ago • ⬇️ 230k • ❤️ 1.24k

NousResearch/Yarn-Llama-2-13b-128k

🗨️ Text Generation • Updated 2 days ago • ⬇️ 1.43k • ❤️ 58

inception-mbzuai/jais-13b-chat

🗨️ Conversational • Updated 6 days ago • ⬇️ 4.5k • ❤️ 62

smallcloudai/Refact-1_6B-fim

🗨️ Text Generation • Updated 5 days ago • ⬇️ 37.8k • ❤️ 51

stabilityai/control-lora

🖼️ Text-to-Image • Updated 18 days ago • ❤️ 494

THUDM/chatglm2-6b

Updated 2 days ago • ⬇️ 2.61M • ❤️ 1.62k

TheBloke/Llama-2-7B-Chat-GGML

🗨️ Text Generation • Updated 1 day ago • ⬇️ 7.78k • ❤️ 488

Waiting for huggingface.co...

<https://huggingface.co/models>

meta-llama/Llama-2-7b

like 2.32k

Text Generation

PyTorch

English

facebook

meta

llama

llama-2

arxiv:2307.09288

Model card

Files and versions

Use with library

Access Llama 2 on Hugging Face

This is a form to enable access to Llama 2 on Hugging Face after you have been granted access from Meta. Please visit the [Meta website](#) and accept our license terms and acceptable use policy before submitting this form. Requests will be processed in 1-2 days.

Your Hugging Face account email address **MUST** match the email you provide on the Meta website, or your request will not be approved.

By agreeing you accept to share your contact information (email and username) with the repository authors.

☐ I agree to share my name, email address and username with Meta and confirm that I have already been granted download access on the Meta website

Submit

Cancel

Llama 2

Llama 2 is a collection of pretrained and fine-tuned generative text models ranging in scale from 7 billion to 70 billion parameters. This is the repository for the 7B pretrained model. Links to other models can be found in the index at the bottom.

Model Details

Note: Use of this model is governed by the Meta license. In order to download the model weights and

Downloads last month

0

Hosted inference API

Text Generation

Inference API has been turned off for this model.

Spaces using meta-llama/Llama-2-7b

11

qiantong-xu/toolbench-leaderboard

AironHeart/llama2

bhaskartripathi/Llama-2-70b-chatbot

mikeee/gradio-chatinterface

Araeynn/llama2

osanseviero/streaming-example

hoyinli/demo-app

realchenyuy/llama2-playground

KingPinX/kbot-1

scp4950/grah

agkbv/meta-llama-Llama-2-7b-hf

https://huggingface.co/meta-llama/Llama-2-7b

Hugging Face Datasets

Hugging Face

Search models, datasets, users...

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Multimodal

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Image-to-Text

Text-to-Video

Visual Question Answering

Graph Machine Learning

Computer Vision

Depth Estimation

Image Classification

Object Detection

Image Segmentation

Image-to-Image

Unconditional Image Generation

Video Classification

Zero-Shot Image Classification

Natural Language Processing

Text Classification

Token Classification

Table Question Answering

Question Answering

Zero-Shot Classification

Translation

Summarization

Conversational

Text Generation

Text2Text Generation

Fill-Mask

Sentence Similarity

Table to Text

Multiple Choice

Text Retrieval

Audio

Text-to-Speech

Automatic Speech Recognition

Audio-to-Audio

Audio Classification

Voice Activity Detection

Datasets61,371

Filter by name

new

Full-text search

Sort: Trending

fka/awesome-chatgpt-prompts

Viewer • Updated Mar 7 • 4.52k • 3.24k

b-mc2/sql-create-context

Viewer • Updated Apr 21 • 2.17k • 114

allenai/dolma

Preview • Updated 18 days ago • 256 • 303

Anthropic/hh-rlhf

Viewer • Updated May 27 • 104k • 634

bigcode/the-stack

Viewer • Updated Apr 13 • 5.9k • 465

F1mc/DISC-Med-SFT

Viewer • Updated 8 days ago • 41 • 12

cerebras/SlimPajama-627B

Viewer • Updated Jul 8 • 693k • 167

shibing624/medical

Viewer • Updated Jun 2 • 462 • 94

timdettmers/openassistant-guanaco

Viewer • Updated May 28 • 25.8k • 197

QingyiSi/Alpaca-CoT

Preview • Updated Jun 19 • 2.2k • 444

dell-research-harvard/AmericanStories

Updated Jun 15 • 426 • 40

garage-bAInd/Open-Platypus

Viewer • Updated 22 days ago • 9.73k • 173

liwu/MNBVC

Viewer • Updated 11 days ago • 1.37k • 211

Open-Orca/OpenOrca

Viewer • Updated 17 days ago • 29.7k • 630

GAIR/lima

Viewer • Updated Jun 8 • 1.52k • 270

ceval/ceval-exam

Viewer • Updated 6 days ago • 140k • 130

databricks/databricks-dolly-15k

Viewer • Updated Jul 1 • 46k • 342

vicgalle/alpaca-gpt4

Viewer • Updated Apr 8 • 2.12k • 86

bigcode/the-stack-dedup

Viewer • Updated 20 days ago • 2.02M • 212

bigcode/starcoderdata

Viewer • Updated May 16 • 17.5k • 160

ps://huggingface.co/datasets?task_categories=task_categories%3Afeature-extraction

https://huggingface.co/datasets

Hugging Face Datasets – b-mc2/sql-create-context

Datasets:

b-mc2/sql-create-context

like 114

Tasks:

Text Generation

Question Answering

Table Question Answering

Languages:

English

Size Categories:

10K<n<100K

Tags:

SQL

code

NLP

+ 5

License:

cc-by-4.0

Dataset card

Files and versions

Community 1

Dataset Viewer

Auto-converted to Parquet API Go to dataset viewer

Split

train (78.6k rows)

Search this dataset

answer (string)	context (string)	question (string)
"SELECT COUNT(*) FROM head WHERE age > 56"	"CREATE TABLE head (age INTEGER)"	"How many heads of the departments are older than 56 ?"
"SELECT name, born_state, age FROM head ORDER BY age"	"CREATE TABLE head (name VARCHAR, born_state VARCHAR, age VARCHAR)"	"List the name, born state and age of the heads of departments ordered by age."
"SELECT creation, name, budget_in_billions FROM department"	"CREATE TABLE department (creation VARCHAR, name VARCHAR, budget_in_billions..."	"List the creation year, name and budget of each department."
"SELECT MAX(budget_in_billions), MIN(budget_in_billions) FROM department"	"CREATE TABLE department (budget_in_billions INTEGER)"	"What are the maximum and minimum budget of the departments?"
"SELECT AVG(num_employees) FROM department WHERE ranking BETWEEN 10 AND 15"	"CREATE TABLE department (num_employees INTEGER, ranking INTEGER)"	"Average number of employees of departments whose rank is between..."
"SELECT name FROM head WHERE born_state <> 'California'"	"CREATE TABLE head (name VARCHAR, born_state VARCHAR)"	"Names of the heads who are not from California state?"
"SELECT DISTINCT T1.creation FROM department AS T1 JOIN management AS T2 ON..."	"CREATE TABLE department (creation VARCHAR, department_id VARCHAR); CREATE..."	"Distinct creation years of departments managed by a secretary..."

< Previous 1 2 3 ... 786 Next >

Downloads last month

2,171

Use in dataset library

Edit dataset card

Train in AutoTrain

Evaluate models

HF Leaderboard

:

Size of downloaded dataset files:

21.8 MB

Size of the auto-converted Parquet files:

6.81 MB

Number of rows:

78,577

Models trained or fine-tuned on b-mc2/sql-crea...

cssupport/t5-small-awesome-text-to-sql

Text2Text Generation • Updated 8 d... • 323 • 6

xtuner/internlm-7b-qlora-sql

Conversational • Updated 26 minutes ago • 10

xianglingjing/llama-2-7b-int4-text-to...

Text Generation • Updated 7 days ago • 9

Hugging Face Spaces

 **Spaces**

Discover amazing ML apps made by the community!

Create new Space or [learn more about Spaces.](#)

new Full-text search Sort: Trending

☆ Spaces of the week 🔥

Running on A100

Seamless M4T

facebook 14 days ago

529

Running on CPU UPGRADE

AI Comic Factory

jbilcke-hf 27 minutes ago

717

Running on A10G

Code Llama 13B Chat

codellama 9 days ago

144

Code Llama Playground

codellama 7 days ago

113

Running on T4

Kosmos 2

ydshieh 13 days ago

119

SEED-Bench Leaderboard

AILab-CVC 6 days ago

20

Running on A10G

AnimateDiff

guoyww 17 days ago

156

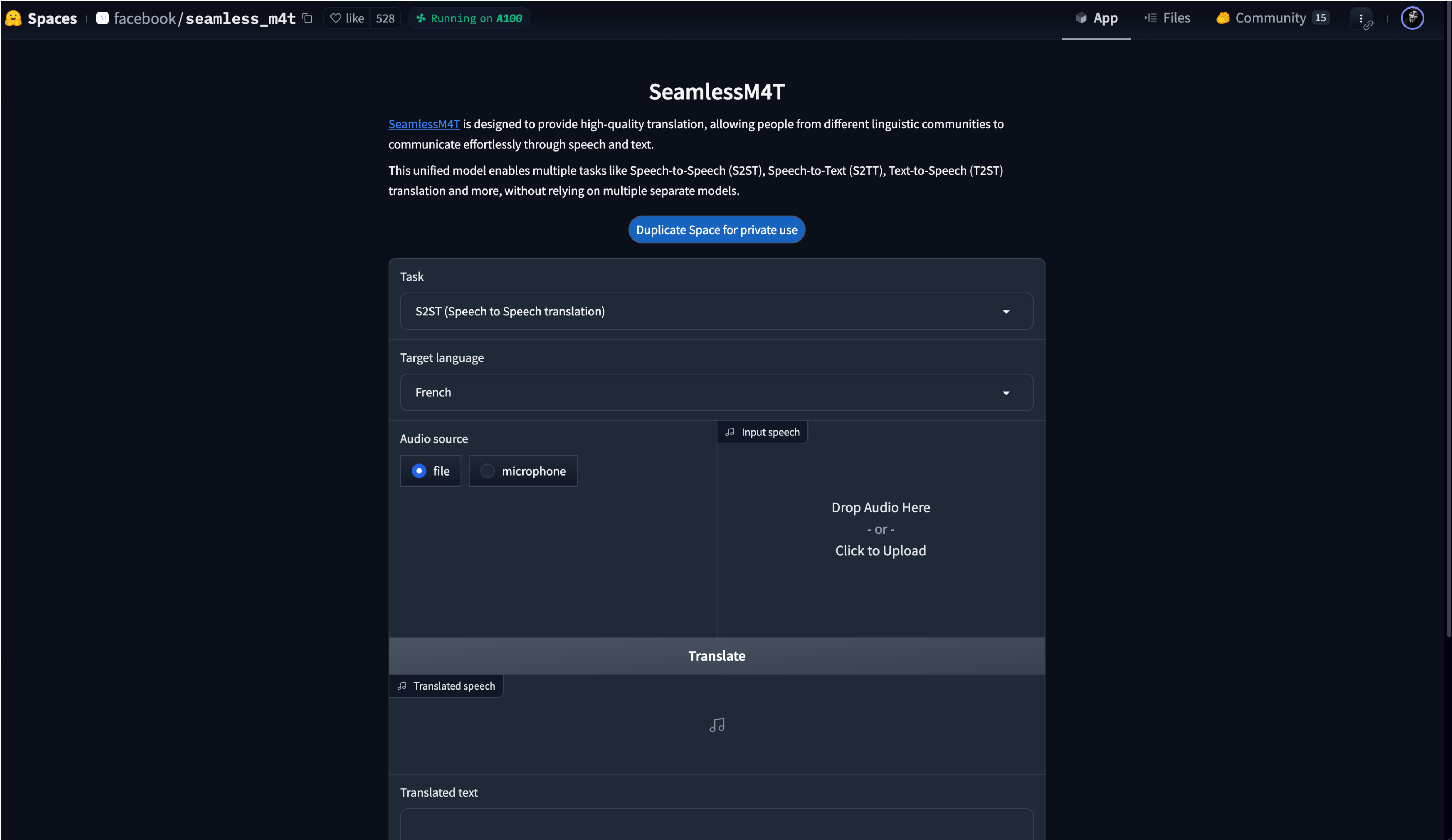
Running on A10G

WavJourney

Audio-AGI 4 days ago

132

Hugging Face Spaces – Facebook/seamless_m4t



 **Documentations**

🔍 Search across all docs

• **Hub**

Host Git-based models, datasets and Spaces on the Hugging Face Hub.

• **Hub Python Library**

Client library for the HF Hub: manage repositories from your Python runtime.

• **Inference API**

Experiment with over 200k models easily using our free Inference API.

• **Accelerate**

Easily train and use PyTorch models with multi-GPU, TPU, mixed-precision.

• **Transformers**

State-of-the-art ML for Pytorch, TensorFlow, and JAX.

• **Datasets**

Access and share datasets for computer vision, audio, and NLP tasks.

• **Huggingface.js**

A collection of JS libraries to interact with Hugging Face, with TS types included.

• **Inference Endpoints**

Easily deploy models to production on dedicated, fully managed infrastructure.

• **Optimum**

Fast training and inference of HF Transformers with easy to use hardware optimization tools.

• **Diffusers**

State-of-the-art diffusion models for image and audio generation in PyTorch.

• **Gradio**

Build machine learning demos and other web apps, in just a few lines of Python.

• **Transformers.js**

Community library to run pretrained models from Transformers in your browser.

• **PEFT**

Parameter efficient finetuning methods for large models

• **AWS Trainium & Inferentia**

Train and Deploy Transformers & Diffusers with AWS Trainium and AWS Inferentia.

 **Hugging Face**

Models

Datasets

Spaces

Docs

• Transformers ▾



V4.33.0 ▾ EN ▾  

GET STARTED

 Transformers

Quick tour

Installation

TUTORIALS

Run inference with pipelines

Write portable code with AutoClass

Preprocess data

Fine-tune a pretrained model

Train with a script

Set up distributed training with  Accelerate

Load and train adapters with  PEFT

Share your model

Agents

Generation with LLMs

TASK GUIDES

 **Transformers**

State-of-the-art Machine Learning for [PyTorch](#), [TensorFlow](#), and [JAX](#).

 Transformers provides APIs and tools to easily download and train state-of-the-art pretrained models. Using pretrained models can reduce your compute costs, carbon footprint, and save you the time and resources required to train a model from scratch. These models support common tasks in different modalities, such as:

 **Natural Language Processing:** text classification, named entity recognition, question answering, language modeling, summarization, translation, multiple choice, and text generation.

 **Computer Vision:** image classification, object detection, and segmentation.

 **Audio:** automatic speech recognition and audio classification.

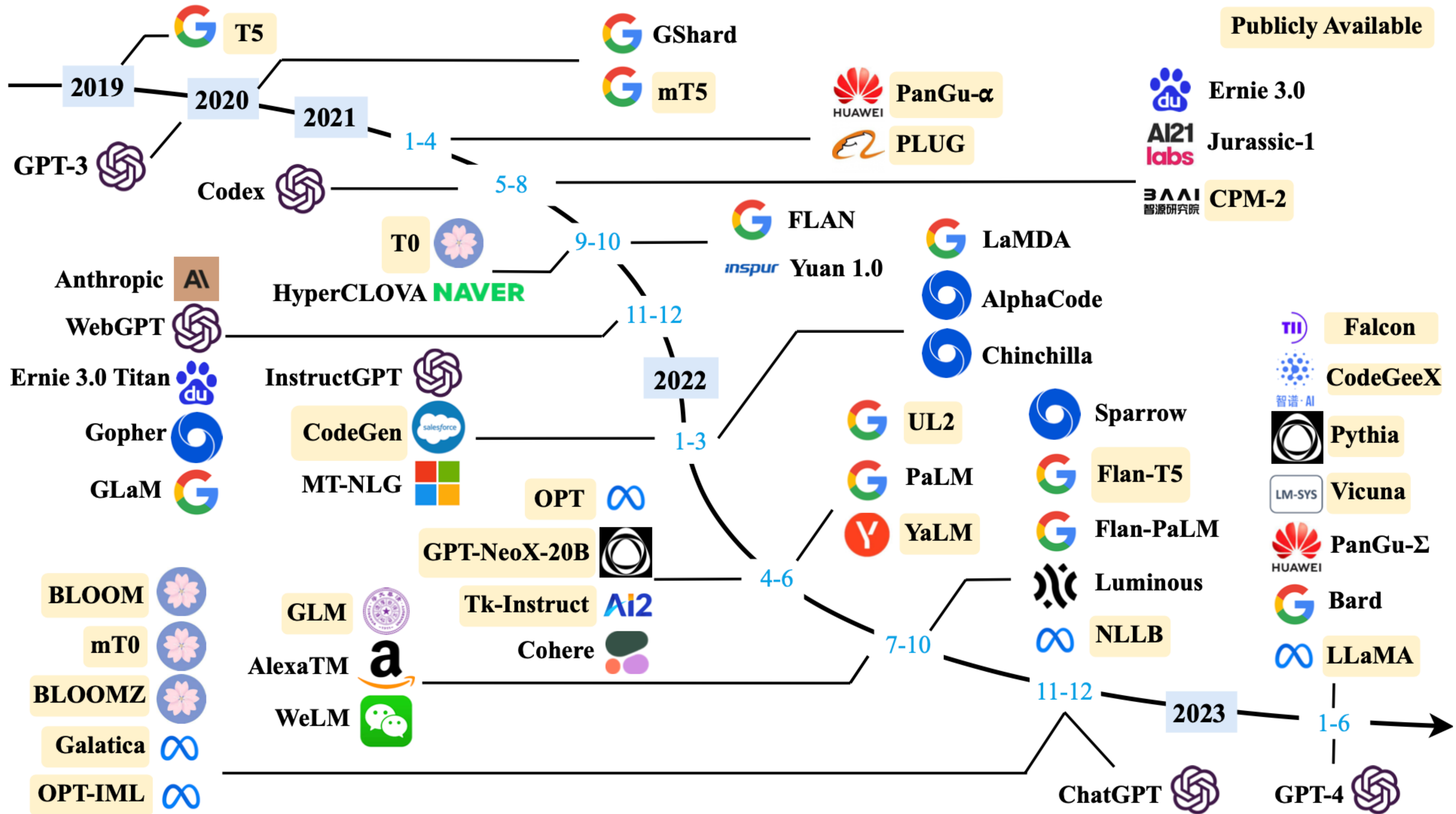
 **Multimodal:** table question answering, optical character recognition, information extraction from scanned documents, video classification, and visual question answering.

 Transformers support framework interoperability between PyTorch, TensorFlow, and JAX. This provides the flexibility to use a different framework at each stage of a model's life; train a model in three lines of code in one framework, and load it for inference in another. Models can also be exported to a format like ONNX and TorchScript for deployment in production environments.

Join the growing community on the [Hub](#), [forum](#), or [Discord](#) today!

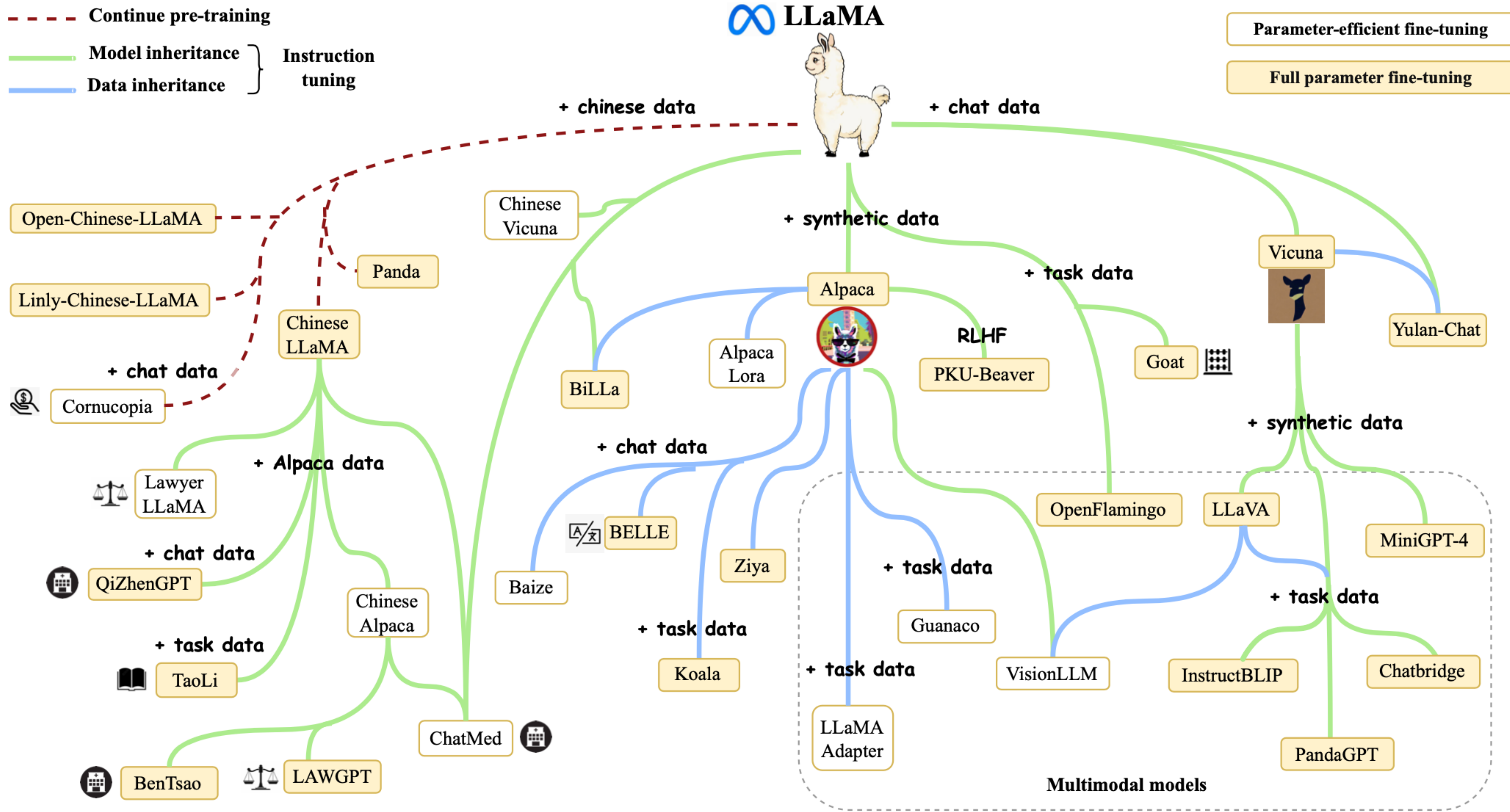
If you are looking for custom support from the Hugging Face team

大模型横向对比



大语言模型 横向对比

	Model	Release Time	Size (B)	Base Model	Adaptation		Pre-train Data Scale	Latest Data Timestamp	Hardware (GPUs / TPUs)	Training Time	Evaluation	
					IT	RLHF					ICL	CoT
Publicly Available	T5 [73]	Oct-2019	11	-	-	-	1T tokens	Apr-2019	1024 TPU v3	-	✓	-
	mT5 [74]	Oct-2020	13	-	-	-	1T tokens	-	-	-	✓	-
	PanGu- α [75]	Apr-2021	13*	-	-	-	1.1TB	-	2048 Ascend 910	-	✓	-
	CPM-2 [76]	Jun-2021	198	-	-	-	2.6TB	-	-	-	-	-
	T0 [28]	Oct-2021	11	T5	✓	-	-	-	512 TPU v3	27 h	✓	-
	CodeGen [77]	Mar-2022	16	-	-	-	577B tokens	-	-	-	✓	-
	GPT-NeoX-20B [78]	Apr-2022	20	-	-	-	825GB	-	96 40G A100	-	✓	-
	Tk-Instruct [79]	Apr-2022	11	T5	✓	-	-	-	256 TPU v3	4 h	✓	-
	UL2 [80]	May-2022	20	-	-	-	1T tokens	Apr-2019	512 TPU v4	-	✓	✓
	OPT [81]	May-2022	175	-	-	-	180B tokens	-	992 80G A100	-	✓	-
	NLLB [82]	Jul-2022	54.5	-	-	-	-	-	-	-	✓	-
	GLM [83]	Oct-2022	130	-	-	-	400B tokens	-	768 40G A100	60 d	✓	-
	Flan-T5 [64]	Oct-2022	11	T5	✓	-	-	-	-	-	✓	✓
	BLOOM [69]	Nov-2022	176	-	-	-	366B tokens	-	384 80G A100	105 d	✓	-
	mT0 [84]	Nov-2022	13	mT5	✓	-	-	-	-	-	✓	-
	Galactica [35]	Nov-2022	120	-	-	-	106B tokens	-	-	-	✓	✓
	BLOOMZ [84]	Nov-2022	176	BLOOM	✓	-	-	-	-	-	✓	-
	OPT-IML [85]	Dec-2022	175	OPT	✓	-	-	-	128 40G A100	-	✓	✓
	LLaMA [57]	Feb-2023	65	-	-	-	1.4T tokens	-	2048 80G A100	21 d	✓	-
	CodeGeeX [86]	Sep-2022	13	-	-	-	850B tokens	-	1536 Ascend 910	60 d	✓	-
	Pythia [87]	Apr-2023	12	-	-	-	300B tokens	-	256 40G A100	-	✓	-
Closed Source	GPT-3 [55]	May-2020	175	-	-	-	300B tokens	-	-	-	✓	-
	GShard [88]	Jun-2020	600	-	-	-	1T tokens	-	2048 TPU v3	4 d	-	-
	Codex [89]	Jul-2021	12	GPT-3	-	-	100B tokens	May-2020	-	-	✓	-
	ERNIE 3.0 [90]	Jul-2021	10	-	-	-	375B tokens	-	384 V100	-	✓	-
	Jurassic-1 [91]	Aug-2021	178	-	-	-	300B tokens	-	800 GPU	-	✓	-
	HyperCLOVA [92]	Sep-2021	82	-	-	-	300B tokens	-	1024 A100	13.4 d	✓	-
	FLAN [62]	Sep-2021	137	LaMDA-PT	✓	-	-	-	128 TPU v3	60 h	✓	-
	Yuan 1.0 [93]	Oct-2021	245	-	-	-	180B tokens	-	2128 GPU	-	✓	-
	Anthropic [94]	Dec-2021	52	-	-	-	400B tokens	-	-	-	✓	-
	WebGPT [72]	Dec-2021	175	GPT-3	-	✓	-	-	-	-	✓	-
	Gopher [59]	Dec-2021	280	-	-	-	300B tokens	-	4096 TPU v3	920 h	✓	-
	ERNIE 3.0 Titan [95]	Dec-2021	260	-	-	-	-	-	-	-	✓	-
	GLaM [96]	Dec-2021	1200	-	-	-	280B tokens	-	1024 TPU v4	574 h	✓	-
	LaMDA [63]	Jan-2022	137	-	-	-	768B tokens	-	1024 TPU v3	57.7 d	-	-
	MT-NLG [97]	Jan-2022	530	-	-	-	270B tokens	-	4480 80G A100	-	✓	-
	AlphaCode [98]	Feb-2022	41	-	-	-	967B tokens	Jul-2021	-	-	-	-
	InstructGPT [61]	Mar-2022	175	GPT-3	✓	✓	-	-	-	-	✓	-
	Chinchilla [34]	Mar-2022	70	-	-	-	1.4T tokens	-	-	-	✓	-
	PaLM [56]	Apr-2022	540	-	-	-	780B tokens	-	6144 TPU v4	-	✓	✓
	AlexaTM [99]	Aug-2022	20	-	-	-	1.3T tokens	-	128 A100	120 d	✓	✓
	Sparrow [100]	Sep-2022	70	-	-	✓	-	-	64 TPU v3	-	✓	-
	WeLM [101]	Sep-2022	10	-	-	-	300B tokens	-	128 A100 40G	24 d	✓	-
	U-PaLM [102]	Oct-2022	540	PaLM	-	-	-	-	512 TPU v4	5 d	✓	✓
	Flan-PaLM [64]	Oct-2022	540	PaLM	✓	-	-	-	512 TPU v4	37 h	✓	✓
	Flan-U-PaLM [64]	Oct-2022	540	U-PaLM	✓	-	-	-	-	-	✓	✓
	GPT-4 [46]	Mar-2023	-	-	✓	✓	-	-	-	-	✓	✓
	PanGu- Σ [103]	Mar-2023	1085	PanGu- α	-	-	329B tokens	-	512 Ascend 910	100 d	✓	-



大模型跑分？

🤗 Open LLM Leaderboard

📌 The 🤗 Open LLM Leaderboard aims to track, rank and evaluate open LLMs and chatbots.

🤗 Submit a model for automated evaluation on the 🤗 GPU cluster on the “Submit” page!

The leaderboard’s backend runs the great [Eleuther AI Language Model Evaluation Harness](#) - read more details in the “About” page!

🏆 LLM Benchmark

📖 About

🚀 Submit here!

Select columns to show

☒ Average 📈

☒ ARC

☒ HellaSwag

☒ MMLU

☒ TruthfulQA

☐ Type

☐ Precision

☐ Hub License

☐ #Params (B)

☐ Hub ❤️

☐ Model sha

☒ Show models removed from the hub

🔍 Search for your model and press ENTER...

Model types

☒ 🟢 pretrained

☒ 🟡 fine-tuned

☒ 🔴 instruction-tuned

☒ 🔵 RL-tuned

Model sizes

☒ < 1.5B

☒ ~3B

☒ ~7B

☒ ~13B

☒ ~35B

☒ 60B+

T	▲	Model	▲	Average 📈	▲	ARC	▲	HellaSwag	▲	MMLU	▲	TruthfulQA	▲
🟡		uni-tianyan/Uni-TianYan 📁		73.81		72.1		87.4		69.91		65.81	
🟡		fangloveskari/ORCA_LLaMA_70B_QLoRA 📁		73.4		72.27		87.74		70.23		63.37	
🟡		garage-bAInd/Platypus2-70B-instruct 📁		73.13		71.84		87.94		70.48		62.26	
🟡		upstage/Llama-2-70b-instruct-v2 📁		72.95		71.08		87.89		70.58		62.25	
🟡		fangloveskari/Platypus_QLoRA_LLaMA_70b 📁		72.94		72.1		87.46		71.02		61.18	
🟡		yeontaek/llama-2-70B-ensemble-v5 📁		72.86		71.16		87.24		69.6		63.45	
🔴		TheBloke/Genz-70b-GPT0 📁		72.82		71.08		87.64		70.26		62.28	
🟡		TheBloke/Platypus2-70B-Instruct-GPT0 📁		72.81		71.25		87.55		69.89		62.54	
🟡		psmathur/model_007 📁		72.72		71.08		87.65		69.04		63.12	
🟡		yeontaek/llama-2-70B-ensemble-v4 📁		72.64		70.9		87.34		69.71		62.6	
🟡		psmathur/orca_mini_v3_70b 📁		72.64		71.25		87.85		70.18		61.27	

Open LLM Leaderboard使用 Eleuther AI语言模型评估工具对模型进行了4个关键基准测试，这是一个统一的框架，用于在大量不同的评估任务上测试生成式语言模型。

- **AI2推理挑战（25-shot）** - 一组小学科学问题。
- **HellaSwag（10-shot）** - 这是一个常识推理测试，对人类来说很容易（约95%），但对SOTA模型来说具有挑战性。
- **MMLU（5-shot）** - 这是一个测量文本模型多任务准确性的测试。该测试涵盖57个任务，包括初等数学、美国历史、计算机科学、法律等等。
- **TruthfulQA（0-shot）** - 这是一个衡量模型倾向于复制在线常见谬误的测试。（注意：Harness中的TruthfulQA实际上是一个6-shots任务，在启动时即使将few-shot示例数量设置为0，也会自动添加6个示例。）

对于所有这些评估，得分越高越好。Open LLM Leaderboard 选择了这些基准作为它们在0-shot和few-shot环境下涵盖了各种推理和广泛知识领域。

HF Open LLM Leaderboard



Spaces: HuggingFaceH4/open_llm_leaderboard

like

4.95k

Running

App

Files

Community 262

main

open_llm_leaderboard

21 contributors

History: 151 commits

+ Contribute

Clémentine

fix bug in metadata display

e0b891a

about 1 hour ago

src

fix bug in metadata display

about 1 hour ago

.gitattributes

1.53 kB

↓

Upload scale-hf-logo.png

3 months ago

.gitignore

166 Bytes

↓

added selector for model type

about 2 months ago

.pre-commit-config.yaml

1.53 kB

↓

Cleaned and refactored the code, improved filtering, added selection of c

12 days ago

Makefile

208 Bytes

↓

Cleaned and refactored the code, improved filtering, added selection of c

12 days ago

README.md

313 Bytes

↓

Adds file to make models backlink to leaderboard

2 days ago

app.py

20.3 kB

↓

fixed gradio issue in #248

2 days ago

models_backlinks.py

44.9 kB

↓

dumps models list in a file

2 days ago

pyproject.toml

548 Bytes

↓

Cleaned and refactored the code, improved filtering, added selection of c

12 days ago

requirements.txt

1.2 kB

↓

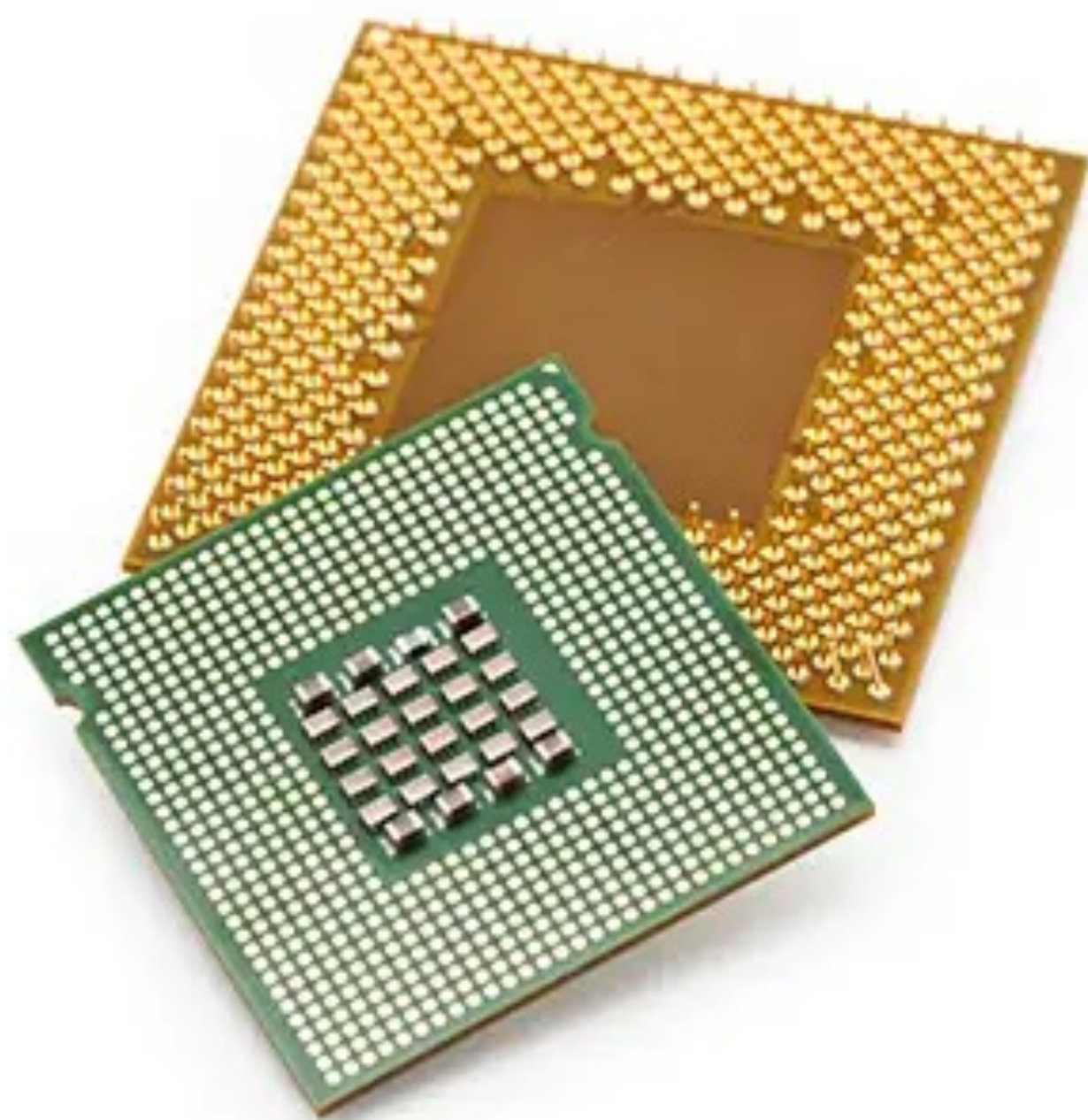
updated requirements to latest transformers release

15 days ago

目录

- 显卡选型推荐指南
 - GPU vs 显卡
 - GPU Core vs AMD CU
 - CUDA Core vs Tensor Core
 - N卡的架构变迁
 - 显卡性能天梯榜

UPGRADE



CPU

OR



GPU

GPU（图形处理单元）：计算机内部的硬件组件，用于处理图形和图像相关的任务。主要功能：

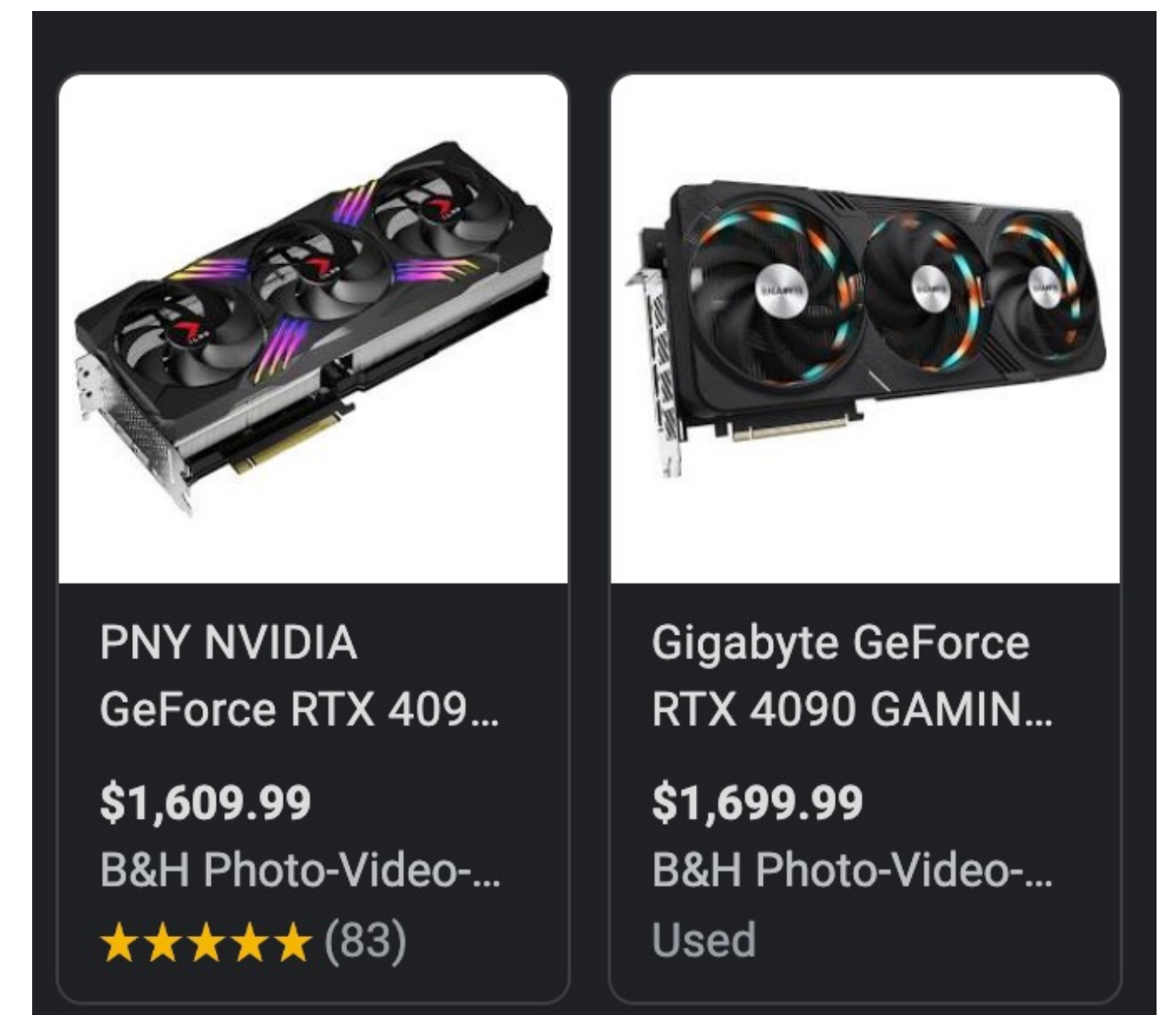
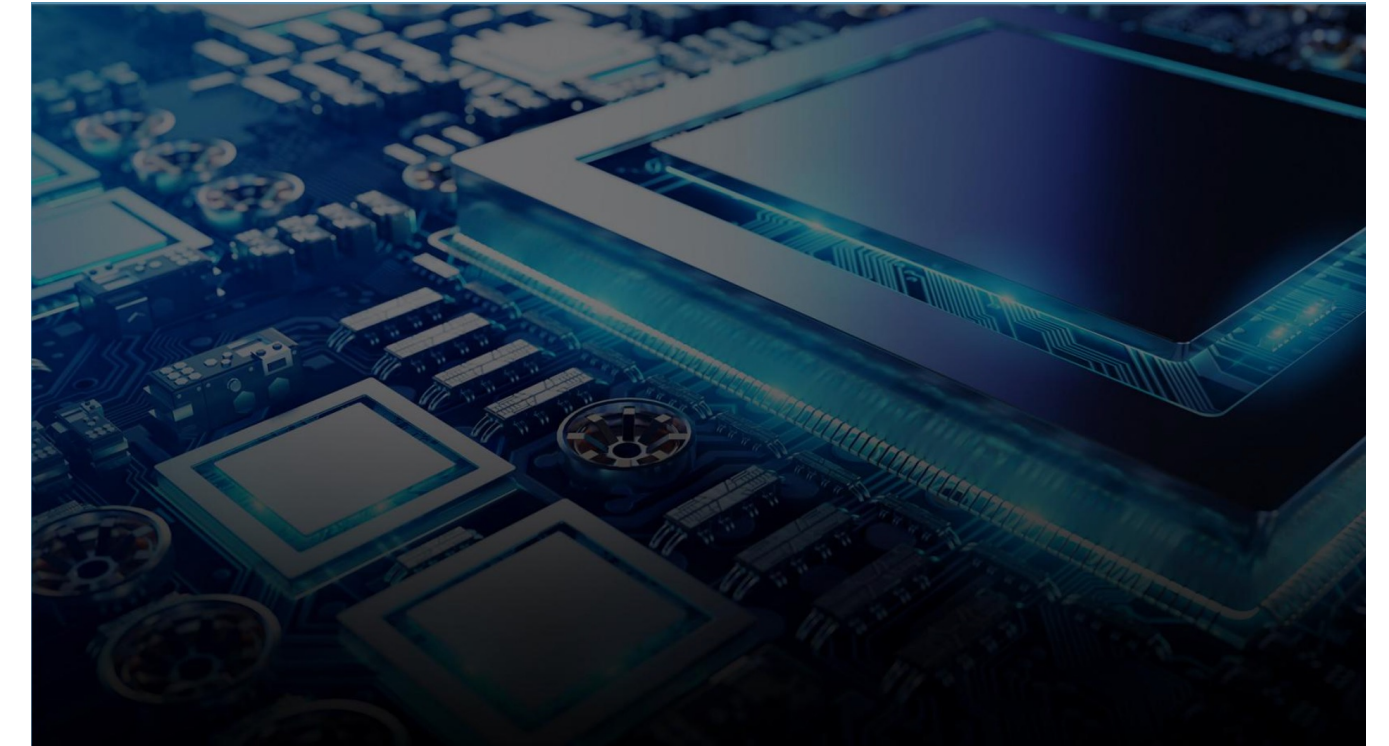
- 处理图形渲染，包括游戏、视频和动画。
- 加速通用计算，如深度学习、科学模拟等。
- 高度并行处理，适用于大规模数据处理和处理密集型任务。

显卡：包含一个或多个GPU芯片的硬件设备，通常用于图形渲染和计算任务。主流显卡的显示芯片主要由NVIDIA（英伟达）和AMD（超威半导体）两大厂商制造，通常将采用NVIDIA显示芯片的显卡称为N卡，而将采用AMD显示芯片的显卡称为A卡。主要作用：

- 将GPU集成到计算机中，通过接口（如PCIe）连接到主板。
- 提供图形输出接口，如HDMI、DisplayPort，使显示器能够显示图像。
- 可以有不同的物理尺寸和散热解决方案，适应各种计算机类型和用途。

GPU与显卡关系：

- GPU是显卡的核心组件，决定了计算性能。
- 显卡则是将GPU与必要的接口、散热、电源管理等元件集成在一起的硬件设备。



显卡虽然是一大块卡，但它真正执行运算的部分只是一枚和 CPU 一样大小的芯片——**显卡核心**；显卡 PCB 上其他组件，都是为它提供电源、通信、存储功能的。

这个小芯片承担着与 CPU 不同的定位和功能，芯片设计思路也完全不同。这就导致，我们谈论 GPU 时，所说的 **Core**，完全不是 CPU 上那个意思。

作为辅助 CPU 执行计算的周边器件，它不需要承担那些系统管理、调度的功能，完全专注于使用（大量的）小核心并行化地执行基础运算（通常都有数千个核心同时执行）。——**GPU Core** 小而且多，这是我们的初步印象。

AMD 倾向于简化整个模型，用 **CU** 来讨论他们的显卡核心数量。**Compute Unit** 指的是一组执行运算的元件的集群（**cluster**），它里面有大量更小的计算单元。

由于技术路线和优势不同，**AMD** 更在乎自己的总线技术（例如 **Infinity Cache**）和集群结构，自然也倾向于以 **CU** 数量来讨论显卡性能。**CU** 不会具体对应到计算单元的数量，而只是跟缓存、总线结构有关。

CU 在概念上更接近 **Nvidia** 的 **SM**。所以千万不要拿 A 卡的 **CU** 与 N 卡的 **CUDA Core** 相比，是两个完全错位的概念。

N 卡上，经常提到的概念是 **CUDA Core**。英伟达倾向于用**最小的运算单元**表示自己的运算能力，**CUDA Core** 指的是一个**执行基础运算的处理元件**。最早的时候，这个「基础组件」类似于 CPU 中负责各种数值计算的那个 **core**，能做很多通用运算。

但后来就不一样了，例如整型计算和浮点型计算单元的数量不一样了。**CUDA Core** 不再能 **1:1** 地对应上这些小运算单元。它的定义就开始变化：有时**2个 CUDA Core** 才能凑出**1个整型运算单元**。

所以 **CUDA Core** 的定义其实是复杂而且变化的。你不能单纯用 **CUDA Core** 的数量来跨代际比较显卡性能（因为它在两代显卡中的定义都不同）。

由于 **CUDA Core** 根本就是为了方便商业营销而创造的概念，英伟达喜欢将其称为「用于营销目的」的核心。随着架构的更替，经过一代又一代的演变。

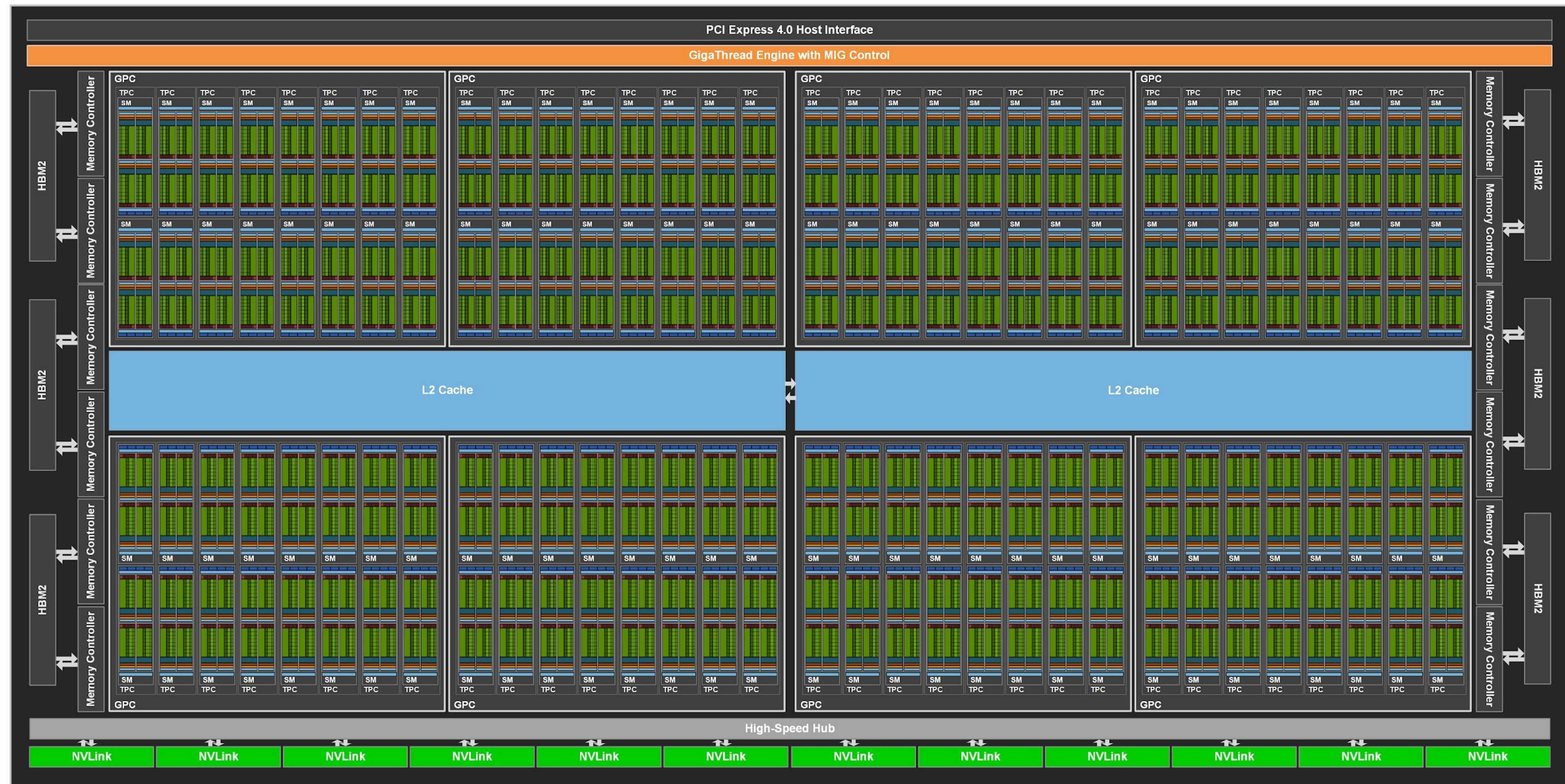
现在 Ampere 架构上所说的 **CUDA Core**（的数量），对应的其实是 **FP32** 计算单元（的数量）

Nvidia A100

Ampere 架构最强显卡 A100 的内部架构：

128 个 **SM**（Streaming Multiprocessor，流式多处理器），密密麻麻地排布在中间蓝色的 L2 Cache 两侧。

关注 **SM** 这个概念，它在 N 卡架构中非常重要，N 卡的多核并行化就是靠平行布置大量的 SM 来实现的。



流式多处理器



机器学习、神经网络的时代到来了。在这种应用场景下，矩阵运算的计算量非常大.....那就干脆.....再来个 **Tensor Core**，专门算矩阵

Tensor Core 有着专门设计的硬件结构，可以把整个矩阵都载入寄存器中批量运算，有十几倍的效率提升。这简直是机器学习神器——自 Volta 架构发布以来，奠定了 N 卡在机器学习领域的江湖地位。

Volta 以来的每一代架构，除了制程上的进步，Nvidia 也不遗余力地更新着他们的 Tensor Core:



N 卡架构变迁

计算能力	架构	发布年代	Cores/SM	总 SM 数	CUDA Cores	L1 Cache (KB)	L2 Cache (KB)
1.0	Tesla						
2.0	Fermi	2009	32	16 SM	512	48	768
3.0	Kepler	2012	192	15 SMX	2880	48	1536
4.0	—						
5.0	Maxwell	2014	128	24 SMM	3072	96	2048
6.0	Pascal	2016	64	60 SM	3840	64	4096
7.0	Volta	2018	64 8 个 Tensor Core	80 SM	5120	与共享内存共用 128 (最多 96)	6144
7.5	Turing	2018	64 8 个 Tensor Core	72 SM	4608	与共享内存共用 128 (最多 96)	6144
8.0	Ampere	2020	64 4 个 Tensor Core	108 SM	6912	与共享内存共用 192 (最多 164)	40960
9.0	Hopper	2022	128 4 个 Tensor Core	144 SM	18432	与共享内存共用 256	61440

Expreview

GPU Rank

All Ranks

Amount of GPUs

317

NVIDIA

54.14%

AMD

45.86%

Changes in decade

Performance



Process(nm)



Latest GPU (click to locate)

Radeon RX 7900 GRE (2023/07/28)

GeForce RTX 4060 (2023/6/28)

EXPreview Presents

Changelog | Suggestions

NVIDIA GPUs

AMD GPUs

超能指数

TDPIW

晶体管数/亿

GeForce RTX 4090 (24G) / 100.0%

80.1% / Radeon RX 7900 XTX (24G)

GeForce RTX 4080 (16G) / 77.8%

70.9% / Radeon RX 7900 XT (20G)

GeForce RTX 3090 Ti (24G) / 69.2%

GeForce RTX 4070 Ti (12G) / 65.3%

GeForce RTX 3090 (24G) / 63.4%

62.6% / Radeon RX 6950 XT (16G)

62.5% / Radeon RX 7900 GRE (16G)

GeForce RTX 3080 Ti (12G) / 62.3%

59.4% / Radeon RX 6900 XT (16G)

GeForce RTX 3080 12GB / 57.7%

GeForce RTX 4070 (12G) / 56.0%

GeForce RTX 3080 (10G) / 55.6%

54.7% / Radeon RX 6800 XT (16G)

47.7% / Radeon RX 6800 (16G)

GeForce RTX 3070 Ti (8G) / 46.1%

GeForce RTX 3070 (8G) / 43.2%

click to compare

Double click to check

基准: GeForce RTX 4090

RANKING

NVIDIA

AMD

HISTORY

TDP

Options

Years 2013-2023 EN

Comparison (click in ranking to compare)

X

Compare

Search (Double click to compare)

Choose or type in here X

Radeon RX 7900 GRE

GeForce RTX 4060

Radeon RX 7600

GeForce RTX 4060 Ti

GeForce RTX 4070

GeForce RTX 4070 Ti

Radeon RX 7900 XT

Radeon RX 7900 XTX

GeForce RTX 4080

Intel Arc A770 16GB

Intel Arc A770

Intel Arc A750

GeForce RTX 3060 8GB

GeForce RTX 4090

Intel Arc A380

Radeon RX 6650 XT

Radeon RX 6750 XT

Radeon RX 6950 XT

Radeon RX 6400

GeForce RTX 3090 Ti

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DjangoPeng add langchain-refactored version of openai-translator1f66fba 17 hours ago67 commits

chatgpt-plugins	add initial version of chatgpt plugins	3 weeks ago
docs	make layout more readable	last month
langchain	add langchain-refactored version of openai-translator	17 hours ago
openai-translator	Merge branch 'main' of github.com:DjangoPeng/openai-quickstart in...	3 weeks ago
openai_api	Remove fine_food_reviews_with_embeddings_1k_demo.csv	last month
.gitignore	add initial version of agents and data	last week
LICENSE	Initial commit	last month
README-CN.md	docs: add 8/13 schedule to readme (#49)	4 days ago
README.md	docs: add 8/13 schedule to readme (#49)	4 days ago

README.md

OpenAI Quickstart

English | 中文

This project is designed as a one-stop learning resource for anyone interested in large language models and their application in Artificial Intelligence Governance and Control (AIGC) scenarios. By providing theoretical foundations, development basics, and hands-on examples, this project offers comprehensive guidance on these cutting-edge topics.

About

A comprehensive guide to understanding and implementing large language models with hands-on examples using LangChain for AIGC applications.

ReadmeApache-2.0 licenseActivity351 stars11 watching171 forks

Releases

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